Table 1: Forecasting variables

Variable	Definition
Section 1: U.S. Business Indicators	
NGDP	Nominal Gross Domestic Product (GDP), levels, seasonally adjusted, annual rate, in billions USD.
PGDP	Gross Domestic Product (GDP) price index, chain weighted, seasonally adjusted, 2010 base year.
CPROF	Level of nominal corporate profits after tax excluding inventory valuation adjustment (IVA) and capital consumption adjustment (CCAdj). Seasonally adjusted, annual rate, in billions USD.
UNEMP	Civilian unemployment rate, levels, seasonally adjusted, percentage points.
EMP	Nonfarm payroll employment, seasonally adjusted, in thousands of jobs.
INDPROD	Industrial production index, seasonally adjusted, index, 2010 base year.
HOUSING	Housing starts, seasonally adjusted, annual rate, in millions.
TBILL	T-Bill rate, 3-month, levels, percentage points.
BOND AAA	AAA corporate bond yield, levels, percentage points.
BOND BAA	BAA corporate bond yield, levels, percentage points.
TBOND	Treasury bond rate, 10-year, levels, percentage points.
Section 2: Real GDP and Its Components	
RGDP	Real Gross Domestic Product, seasonally adjusted, annual rate, 2010 base year.
RCONSUM	Real personal consumption expenditures, seasonally adjusted, annual rate, 2010 base year.
RNRESIN	Real nonresidential fixed investment, seasonally adjusted, annual rate, 2010 base year.
RRESINV	Real residential fixed investment, seasonally adjusted, annual rate, 2010 base year.
RFEDGOV	Real federal government consumption and gross investment, seasonally adjusted, annual rate, 2010 base year.
RSLGOV	Real state and local government consumption and gross investment, seasonally adjusted, annual rate, 2010 base year.
RCBI	Real change in private inventories, levels, chain-weighted real change in private inventories. Seasonally adjusted, annual rate, 2010 base year.
REXPORT	Real net exports of goods and services, levels, chain-weighted real net exports. Seasonally adjusted, annual rate, 2010 base year.
Section 3: CPI and PCE Inflation	
CPI	Headline CPI inflation rate, seasonally adjusted, annual rate, percentage points.
CORECPI	Core CPI inflation rate, seasonally adjusted, annual rate, percentage points.
PCE	PCE inflation rate, headline chain-weighted, seasonally adjusted, annual rate, percentage points.
COREPCE	Core PCE inflation rate, core chain-weighted PCE inflation rate, seasonally adjusted, annual rate, percentage points.

Table 2: Definitions of forecaster characteristics

Characteristic	Description
Name	Full name of the forecaster as listed in the SPF acknowledgement panel.
Gender (inferred)	Gender inferred based on publicly available information (e.g., name, social media).
Affiliation	The institution or company the forecaster is associated with (e.g., university, bank).
Affiliation type	Type of organization the forecaster works for, such as academic, financial, forecasting center, government, etc.
Position in given year	The forecaster's job title or role when the forecast was made (e.g., professor, economist).
Highest degree earned	The highest degree the forecaster has earned (e.g., Bachelor, Master, PhD).
Degree area/field	The field of study for the forecaster's degree (e.g., Economics, Finance).
Graduation year	The year the forecaster completed their highest degree, used to estimate experience and age.
Alma mater	The institution where the forecaster earned their highest degree.
Company location	The geographic location of the forecaster’s affiliated institution or company.
Gives interviews	Whether the forecaster gives public interviews or engages with media (yes/no).
Has Twitter	Whether the forecaster has a public Twitter profile (yes/no).

Table 3: Descriptive statistics of forecast variables

Variable	N	Mean	SD	Skewness	Kurtosis	Median	Min	Max	ACF(1)
Section 1: U.S. Business Indicators									
Nominal GDP (x1000)	5308	16.03	4.72	0.55	-0.44	15.01	8.68	27.62	0.90
GDP Price	5308	92.22	13.34	0.41	-0.37	91.40	70.60	122.85	0.90
Corporate Profits (x1000)	5308	1.38	0.57	0.52	-0.31	1.27	0.49	2.90	0.87
Unemployment Rate	5308	5.75	2.04	1.40	2.32	5.00	3.40	14.70	0.71
Non-Farm Payroll (x1000)	5308	138.61	8.04	0.72	-0.71	136.21	127.68	156.92	0.88
Industrial Production	5308	110.36	13.76	1.34	0.96	106.00	92.50	147.30	0.86
Housing Starts	5308	1.29	0.44	-0.13	-1.01	1.25	0.51	2.11	0.86
Treasury Bill Rate (3M)	5308	1.88	2.00	0.76	-0.91	1.15	0.02	6.29	0.83
AAA Corp Bond Yield	5308	4.83	1.34	0.25	-0.53	4.93	2.14	7.78	0.88
BAA Corp Bond Yield	5308	5.88	1.42	0.17	-0.79	6.01	3.24	8.88	0.85
Treasury Bond Rate (10Y)	5308	3.35	1.37	0.16	-0.79	3.39	0.62	6.66	0.84
Section 2: Real GDP and Its Components									
Real GDP (x1000)	5308	14.05	3.78	0.29	-1.06	13.33	7.68	22.49	0.90
Real PCE (x1000)	5308	9.81	2.62	0.27	-0.98	9.42	5.25	15.49	0.91
Real Non-Res Fixed Inv (x1000)	5308	1.82	0.65	0.68	-1.01	1.44	0.99	3.27	0.90
Real Res Fixed Inv	5308	493.98	122.31	0.10	-1.27	504.30	293.30	734.00	0.86
Real Federal C&GI (x1000)	5308	0.97	0.28	-0.11	-1.18	1.04	0.45	1.49	0.90
Real State/Local C&GI (x1000)	5308	1.52	0.37	0.19	-1.07	1.46	0.85	2.34	0.89
Real Change in Private Inv	5308	24.95	69.41	-1.12	3.69	34.50	-287.00	193.20	0.37
Real Net Exports	5308	-610.58	285.40	-1.43	1.27	-533.10	-1544.70	-250.00	0.86
Section 3: CPI and PCE Inflation									
CPI Inflation Rate	5308	2.52	3.14	-1.14	7.41	2.48	-13.39	10.73	0.00
Core CPI Inflation Rate	5308	2.33	1.49	2.21	8.36	2.19	-1.09	10.21	0.31
PCE Inflation Rate	5308	0.69	7.61	-4.05	16.31	2.14	-40.91	6.91	0.03
Core PCE Inflation Rate	5308	-0.21	2.95	-9.29	85.64	0.03	-28.31	2.00	-0.04

Table 4: Forecast accuracy of human and AI-generated SPF forecasts

	Horizon 0			Horizon 1			Horizon 4		
Variable	**AI**	**Human**	**P-val**	**AI**	**Human**	**P-val**	**AI**	**Human**	**P-val**
Section 1: U.S. Business Indicators									
Nominal GDP	**44.39**	187.68	0.00***	**152.96**	178.49	0.00***	426.85	**381.28**	0.00***
GDP Price	**22.02**	22.07	0.30 [0.91]	**22.01**	22.12	0.41 [0.77]	**21.97**	22.27	0.62 [0.32]
Corporate Profits	**46.59**	71.52	0.24 [0.68]	**98.86**	101.25	0.56 [0.95]	**182.94**	186.90	0.70 [0.50]
Unemployment Rate	0.53	**0.38**	0.81 [0.27]	0.73	**0.59** [0.58]	0.97 [0.99]	1.05	**0.94**	0.69 [0.03**]
Non-Farm Payroll	984.75	**465.16**	0.06* [0.08*]	1487.62	**805.58**	0.21 [0.01**]	2867.16	**1952.10**	0.27 [0.08*]
Industrial Production	2.33	**1.64**	0.18 [0.00***]	3.53	**2.98**	0.17 [0.00***]	**5.90**	6.32 [6.24]	0.00***
Housing Starts	0.10	**0.09**	0.44 [0.02**]	0.13	**0.12**	0.24 [0.00***]	**0.21**	**0.21**	0.07* [0.02**]
Treasury Bill Rate (3M)	0.47	**0.26**	0.95 [0.85]	0.67	**0.44**	0.71 [0.50]	1.29	**1.21**	0.12 [0.00***]
AAA Corp Bond Yield	0.41	**0.28**	0.99 [0.74]	0.48	**0.44**	0.63 [0.26]	**0.71**	0.73	0.14 [0.00***]
BAA Corp Bond Yield	0.46	**0.30**	0.03** [0.00***]	0.56	**0.49**	0.02** [0.00***]	**0.80**	0.81	0.00***
Treasury Bond Rate (10Y)	0.47	**0.32**	0.85 [0.67]	0.57	**0.51**	0.70 [0.29]	**0.83**	0.88	0.03** [0.00***]
Section 2: Real GDP and Its Components									
Real GDP	173.30	**126.26**	0.00***	314.34	**209.62**	0.02** [0.00***]	650.31	**570.78**	0.00***
Real PCE	**61.41**	90.67	0.00***	166.62	**130.56**	0.01*** [0.00***]	395.96	**332.31**	0.00***
Real Non-Res Fixed Inv	**20.51**	25.91	0.03** [0.00***]	54.78	**48.91**	0.06* [0.00***]	**131.69**	134.24	0.01*** [0.00***]
Real Res Fixed Inv	**10.10**	10.49	0.78 [0.76]	25.86	**18.12**	0.76 [0.71]	59.55	**54.28**	0.76 [0.65]
Real Federal C&GI	12.63	**6.79**	0.84 [0.87]	29.70	**19.39**	0.65 [0.63]	57.52	**45.84**	0.50 [0.43]
Real State/Local C&GI	12.72	**8.10**	0.02** [0.00***]	31.43	**24.47**	0.15 [0.00***]	78.72	**70.18**	0.12 [0.00***]
Real Change in Private Inv	**14.26**	24.94	0.49 [0.27]	42.52	**38.14**	0.51 [0.15]	56.55	**49.10**	0.39 [0.06*]
Real Net Exports	21.17	**16.55** [16.82]	0.29 [0.14]	53.99	**43.31** [42.76]	0.26 [0.00***]	114.92	**91.89** [91.86]	0.05* [0.00***]
Section 3: CPI and PCE Inflation									
CPI Inflation Rate	2.40	**1.99**	0.93 [0.77]	2.24	**2.14**	0.74 [0.09*]	**2.04**	2.06	0.87 [0.22]
Core CPI Inflation Rate	0.90	**0.82**	0.59 [0.17]	**0.81**	0.87	0.30 [0.01***]	1.02	**1.00**	0.37 [0.01**]
PCE Inflation Rate	**1.93**	2.49	0.75 [0.57]	**2.64**	2.73	0.75 [0.56]	3.16	**3.14**	0.89 [0.88]
Core PCE Inflation Rate	2.43	**2.31**	0.80 [0.03**]	2.37	**2.21**	0.94 [0.30]	2.19	**2.15**	0.66 [0.12]

Table 5: Percentage survey quarters where AI is more accurate than the human SPF

	Horizon 0		Horizon 1		Horizon 4	
Variable	**Share**	**P-val**	**Share**	**P-val**	**Share**	**P-val**
Section 1: U.S. Business Indicators						
Nominal GDP	**0.94**	0.00***	**0.70**	0.00***	**0.55**	0.32
GDP Price	**0.80**	0.00***	**0.86**	0.00***	**0.88**	0.00***
Corporate Profits	**0.64**	0.01***	**0.58**	0.13 [0.14]	**0.55**	0.32 [0.33]
Unemployment Rate	0.33 [0.34]	0.02** [0.03**]	0.37	0.10 [0.12]	0.41	0.17
Non-Farm Payroll	0.07	0.00***	0.10	0.00***	0.20	0.00***
Industrial Production	0.33	0.00***	0.41	0.07*	**0.60**	0.06*
Housing Starts	0.49	0.92	0.45	0.37	**0.51**	0.92
Treasury Bill Rate (3M)	0.24	0.00***	0.28	0.00***	0.50	1.00
AAA Corp Bond Yield	0.23	0.00***	0.46	0.49 [0.50]	**0.55**	0.42
BAA Corp Bond Yield	0.23	0.00***	0.39	0.11	**0.55**	0.46 [0.45]
Treasury Bond Rate (10Y)	0.34	0.00***	0.46	0.51	**0.57**	0.20
Section 2: Real GDP and Its Components						
Real GDP	**0.63**	0.01**	**0.55**	0.32	0.50	1.00
Real PCE	**0.86**	0.00***	**0.51**	0.92	0.34	0.00***
Real Non-Res Fixed Inv	**0.70**	0.00***	**0.52**	0.76	**0.56**	0.22 [0.23]
Real Res Fixed Inv	**0.69**	0.00***	0.39	0.04**	0.43	0.16
Real Federal C&GI	**0.76**	0.00***	0.42	0.11	0.36	0.01***
Real State/Local C&GI	**0.59**	0.09* [0.08*]	0.37	0.01**	0.38	0.02**
Real Change in Private Inv	**0.74**	0.00***	0.47	0.62	0.40	0.06*
Real Net Exports	0.49 [0.50]	0.92 [1.00]	0.44	0.27	0.29	0.00***
Section 3: CPI and PCE Inflation						
CPI Inflation Rate	0.45	0.38	0.48	0.80	**0.55**	0.54
Core CPI Inflation Rate	0.41	0.19	**0.62**	0.16 [0.15]	0.37	0.07* [0.08*]
PCE Inflation Rate	**0.75**	0.00***	**0.63**	0.08*	**0.57**	0.35
Core PCE Inflation Rate	0.50	1.00	0.44	0.46	**0.55**	0.53

Table 6: Forecast accuracy of best human and AI SPF forecasters

	Horizon 0			Horizon 1			Horizon 4		
Variable	**AI**	**Human**	**P-val**	**AI**	**Human**	**P-val**	**AI**	**Human**	**P-val**
Section 1: U.S. Business Indicators									
Nominal GDP	**16.04**	115.74	0.00***	**49.00**	165.42	0.09* [0.08*]	**122.43**	304.78	0.02**
GDP Price	**4.88**	7.31	0.05**	**3.73**	6.13	0.04**	**0.65**	2.20	0.05**
Corporate Profits	**9.23**	55.20	0.02**	**19.67**	66.44	0.06*	**47.47**	125.77	0.03**
Unemployment Rate	**0.06**	**0.06**	0.65 [0.64]	**0.12**	0.18	0.22	**0.12**	0.44	0.10 [0.10*]
Non-Farm Payroll	328.08	**85.00**	1.00	850.00	**45.00**	0.47	1971.00	**121.00**	0.87
Industrial Production	**0.32**	0.92	0.37 [0.36]	**0.58**	1.23	0.33 [0.32]	**1.50**	2.43	0.05**
Housing Starts	**0.03**	0.04	0.66	**0.05**	0.07	0.28 [0.27]	**0.03**	0.28	0.02**
Treasury Bill Rate (3M)	0.03	**0.02**	0.63 [0.67]	0.04	**0.03**	0.39	0.09	**0.07**	0.60
AAA Corp Bond Yield	0.21	**0.16**	0.96	**0.15**	0.21	0.79 [0.80]	0.24	**0.15**	0.59
BAA Corp Bond Yield	0.66	**0.27**	0.56	0.91	**0.39**	0.19	**0.25**	0.35	0.18
Treasury Bond Rate (10Y)	**0.09**	0.15	0.99	**0.12**	0.16	0.23 [0.24]	**0.17**	0.50	0.15
Section 2: Real GDP and Its Components									
Real GDP	**39.12**	103.89	0.04**	**56.80**	110.74	0.03**	**55.70**	272.73	0.01***
Real PCE	**5.80**	50.58	0.00***	**34.83**	53.36	0.14	93.62	**53.36**	0.35
Real Non-Res Fixed Inv	**6.39**	28.26	0.00***	**33.66**	41.69	0.57 [0.56]	**49.35**	100.42	0.02**
Real Res Fixed Inv	**1.10**	2.64	0.25 [0.24]	**7.89**	13.19	0.36 [0.35]	24.71	**23.87**	0.12
Real Federal C&GI	**1.60**	2.01	0.96	12.51	**6.03**	0.82	20.63	**13.27**	0.74 [0.75]
Real State/Local C&GI	**4.29**	5.85	0.00***	**6.35**	9.46	0.02**	**14.04**	17.78	0.05* [0.05**]
Real Change in Private Inv	**2.88**	21.49	0.03**	**8.03**	19.96	0.66 [0.67]	**17.38**	36.48	0.20
Real Net Exports	**8.54**	15.16	0.12	16.00	**10.80**	0.04**	18.48	**14.40**	0.92
Section 3: CPI and PCE Inflation									
CPI Inflation Rate	0.94	**0.87**	0.88 [0.89]	**0.46**	0.87	0.71 [0.69]	**0.32**	0.34	1.00
Core CPI Inflation Rate	0.68	**0.19**	0.87	0.70	**0.25**	0.88	**0.38**	**0.38**	0.74
PCE Inflation Rate	**0.21**	0.54	1.00	**0.35**	2.08	0.17	**0.33**	0.71	0.73 [0.74]
Core PCE Inflation Rate	1.21	**1.10**	0.52 [0.51]	1.28	**1.14**	0.34 [0.33]	0.70	**0.59**	0.42

Table 7: Cross-Sectional Disagreement by Forecast Horizon

Variable	σh+1	σh+4	Ratio	t-stat	N
Panel A: Human					
CPI	0.972	0.701	0.72	-6.82***	100
UNEMP	0.155	0.356	2.29	20.98***	100
RGDP	40.16	99.90	2.49	14.54***	100
TBILL	0.126	0.363	2.88	18.11***	100
Panel B: AI					
CPI	0.266	0.253	0.95	-0.64	94
UNEMP	0.075	0.115	1.53	4.88***	72
RGDP	122.3	171.0	1.40	9.14***	91
TBILL	0.069	0.116	1.68	6.28***	73

Table 8: Relative accuracy of AI forecasts with fewer prompt inputs

	Generic		No real-time data		No real-time or past SPF data		Recall
Variable	**h0**	**h4**	**h0**	**h4**	**h0**	**h4**	**h0**
Section 1: U.S. Business Indicators							
Nominal GDP	**2.21**	**1.02**	**3.92**	0.95	**6.32***** [6.32]	**1.16***** [1.16]	**89.03*****
GDP Price	0.97	**1.01**	0.97	**1.01**	0.72***	0.77***	**1.09*****
Corporate Profits	**1.74**	**1.01**	**2.84** [2.84***]	**1.17**	**6.20** [6.20***]	**2.25*****	**11.64**
Unemployment Rate	0.98* [0.98]	**1.01**	**1.04***** [1.04**]	**1.04**	0.66	0.90***	0.61
Non-Farm Payroll	**1.22**	**1.01** [1.01**]	**1.59** [1.59***]	**1.35***** [1.35]	**1.35*** [1.35]	**2.57*****	**10.91*****
Industrial Production	**1.05**	1.00	**1.36***** [1.36]	**1.22**	**3.66*****	**1.18*****	**3.13** [3.13***]
Housing Starts	**1.04**	**1.04**** [1.04]	**1.24***** [1.24]	**1.05**	**1.01**	**1.60*****	**10.49*****
Treasury Bill Rate (3M)	**1.03**	**1.01**	**1.06**	**1.03**	0.72	0.94*** [0.94]	0.68*** [0.68]
AAA Corp Bond Yield	**1.04**	**1.02**	**1.03**	**1.12**	**1.97*****	0.97***	**1.65*****
BAA Corp Bond Yield	**1.04**	**1.05**	**1.06**	**1.14**	**1.88** [1.88***]	1.00 [1.00***]	**1.51*****
Treasury Bond Rate (10Y)	**1.01**	**1.01**	1.00	**1.10***** [1.10]	0.71	**1.02**	0.55
Section 2: Real GDP and Its Components							
Real GDP	**1.39**	**1.03**	**1.44**	1.00	**5.96*****	**1.73*****	**20.15*****
Real PCE	**1.21**	0.98* [0.98]	**2.33** [2.33***]	0.96	**12.06*****	**2.09*****	**41.67*****
Real Non-Res Fixed Inv	**1.02**	**1.01**	**2.38*****	**1.12**	**22.04*****	**4.29*****	**35.33*****
Real Res Fixed Inv	0.87*** [0.87]	1.00	**2.05**	**1.06**	**12.10*****	**3.66*** [3.66***]	**19.08***** [19.08**]
Real Federal C&GI	**1.37**	0.95	**2.51**** [2.51*]	**1.04**	**20.68*****	**5.15*****	**56.83*****
Real State/Local C&GI	**2.60** [2.60***]	0.95	**3.46** [3.46***]	**1.05**	**27.36*****	**4.72*****	**48.63***** [48.63]
Real Change in Private Inv	0.98	0.96 [0.96*]	**2.59***** [2.59]	0.87	**3.73*****	**1.29*****	**7.27*****
Real Net Exports	1.00 [1.00***]	0.95	**2.28**	0.93	**6.62*****	**1.39***** [1.39*]	**30.44**** [30.44***]
Section 3: CPI and PCE Inflation							
CPI Inflation Rate	0.90*** [0.90]	**1.02**	0.91*** [0.91]	**1.02** [1.02***]	0.89*** [0.89*]	**1.02*****	0.85* [0.85***]
Core CPI Inflation Rate	0.94	**1.05**	0.99	**1.08**	0.93	**1.05** [1.05***]	0.80
PCE Inflation Rate	**1.26**	1.00	**1.40**	0.91 [0.91***]	**1.34**	1.00	**1.27**
Core PCE Inflation Rate	0.98 [0.98**]	**1.01**	0.97** [0.97***]	**1.05**	0.98	**1.14*****	**1.05**
Average	**1.21**	1.00	**1.76**	**1.06**	**6.08**	**1.87**	**17.16**

Table 9: AI Forecast Performance During Unexpected Events

Event	Event date (days after survey deadline)	Relevant variables	Average Z-score	
			Relevant variables Z-score	All variables Z-score
9/11 Attacks	2001-09-11 (27 days)	RGDP, UNEMP, CPI, TBILL, TBOND	0.28	0.01
Lehman Brothers Collapse	2008-09-15 (39 days)	RGDP, UNEMP, CPI, TBILL, TBOND, HOUSING	1.49	0.49
Flash Crash	2010-05-06 (-5 days)	RGDP, CPI	0.00	-0.12
S&P Downgrade of U.S. Debt	2011-08-05 (-3 days)	TBILL, TBOND, RGDP	0.26	-0.06
COVID-19 Pandemic Declaration	2020-03-11 (29 days)	RGDP, UNEMP, CPI, HOUSING, INDPROD	1.19	0.77
Russia-Ukraine War	2022-02-24 (17 days)	RGDP, CPI, TBILL, TBOND	0.72	0.29
Silicon Valley Bank Collapse	2023-03-10 (31 days)	RGDP, UNEMP, TBILL, TBOND	-0.20	-0.09

Table 10: AI Forecast Error Regressions on Knowledge Cut-Off Dummy Variable

	RGDP	UNEMP	CPI	TBILL
Intercept	366.06***	0.714***	2.135***	0.652***
	(65.43)	(0.141)	(0.210)	(0.076)
I(t > Oct 2023)	715.56*	-0.539	-1.323	-0.137
	(340.01)	(0.732)	(1.091)	(0.397)
Observations	108	108	108	108
R^2	0.040	0.005	0.014	0.001

Table 11: Forecast Revision Regressions: Human vs AI

	Error Correction				Consensus Tracking			
	RGDP	CPI	UNEMP	TBILL	RGDP	CPI	UNEMP	TBILL
Panel A: Human								
Lagged Error(i,t−1)	0.981*** [1.056***]	0.881*** [0.328***]	0.956*** [1.718***]	0.975*** [1.638***]				
	(0.008) [(0.012)]	(0.016) [(0.023)]	(0.007) [(0.072)]	(0.008) [(0.055)]				
Distance from Consensus(i,t−1)					-0.612*** [-0.871***]	-0.690*** [-0.741***]	-0.868*** [-1.122***]	-0.296*** [-0.724***]
					(0.134) [(0.159)]	(0.035) [(0.043)]	(0.064) [(0.322)]	(0.102) [(0.089)]
Observations	3,437 [3,296]	3,452 [3,179]	3,615 [3,287]	3,309 [3,083]	3,437 [3,296]	3,452 [3,179]	3,615 [3,287]	3,309 [3,083]
R^2	0.941 [0.751]	0.636 [0.094]	0.911 [0.281]	0.922 [0.421]	0.015 [0.016]	0.140 [0.164]	0.072 [0.048]	0.004 [0.047]
Panel B: AI								
Lagged Error(i,t−1)	0.610*** [0.621***]	0.443*** [0.654***]	0.647*** [1.141***]	0.594*** [0.957***]				
	(0.028) [(0.025)]	(0.018) [(0.008)]	(0.008) [(0.019)]	(0.008)				
Distance from Consensus(i,t−1)					-1.016*** [-1.009***]	-0.693*** [-1.086***]	-0.510*** [-0.952***]	-1.143*** [-1.090***]
					(0.004) [(0.001)]	(0.017) [(0.009)]	(0.049) [(0.052)]	(0.011) [(0.006)]
Observations	5,188 [5,059]	5,188 [5,059]	5,188 [5,059]	5,188 [5,059]	5,188 [5,059]	5,188 [5,059]	5,188 [5,059]	5,188 [5,059]
R^2	0.568 [0.625]	0.354 [0.823]	0.946 [0.290]	0.827 [0.883]	0.776 [0.991]	0.551 [0.817]	0.086 [0.153]	0.716 [0.867]

Table 12: Reasoning Categories by Economic Period

		Detailed Categories				ECB Aggregation	
Period	**N**	**Mechanical**	**Policy**	**Model**	**Event**	**Model-Based**	**Judgment-Based**
Normal	4,626	98.6%	55.2%	86.7%	19.7%	99.6%	63.3%
Financial Crisis	346	84.7%	34.4%	37.6%	82.4%	88.4%	87.6%
COVID-19	204	72.1%	59.8%	26.5%	77.0%	72.5%	79.9%
Inflation Shock	212	85.4%	87.3%	97.2%	30.2%	98.1%	87.7%

Table B.13: Out-of-sample forecast accuracy (mean absolute errors) of SPF and AI-generated SPF forecasts

	Horizon 0			Horizon 1			Horizon 2 [4]		
Variable	**AI**	**Human**	**P-val**	**AI**	**Human**	**P-val**	**AI**	**Human**	**P-val**
Section 1: U.S. Business Indicators									
Nominal GDP	725.96	**255.92**	0.03**	844.98	**157.99**	0.03**	789.59	**573.08**	0.09* [0.10*]
GDP Price	2.64	**2.62**	0.38	3.24	**3.13**	0.51 [0.50]	5.10	**5.06**	0.58
Corporate Profits	429.19	**225.32**	0.10 [0.11]	374.94	**335.66**	0.24 [0.26]	629.90	**569.84**	0.68 [0.69]
Unemployment Rate	0.17	**0.10**	0.63	0.19	**0.18**	0.01**	0.25	**0.17**	0.02**
Non-Farm Payroll	463.79	**334.69**	0.08* [0.09*]	1051.89	**530.01**	1.00	3477.57	**1220.57**	0.98 [0.99]
Industrial Production	0.62	**0.57**	0.32	**0.80**	0.81	0.08*	1.61	**0.76**	0.31 [0.32]
Housing Starts	0.07	**0.04**	0.01***	0.11	**0.06**	0.01***	0.22	**0.07**	0.01** [0.02**]
Treasury Bill Rate (3M)	0.45	**0.19**	0.58	0.53	**0.23**	0.75 [0.76]	0.56	**0.35**	0.02** [0.01**]
AAA Corp Bond Yield	0.47	**0.19**	0.33 [0.34]	**0.36**	0.40	0.35	**0.32**	0.34	0.92
BAA Corp Bond Yield	0.50	**0.20**	0.42	0.48	**0.35**	0.28 [0.27]	0.35	**0.29**	0.77
Treasury Bond Rate (10Y)	0.42	**0.26**	0.74 [0.73]	**0.36**	0.44	0.78	**0.31**	0.48	0.24 [0.23]
Section 2: Real GDP and Its Components									
Real GDP	1019.93	**289.94**	0.01***	1341.85	**557.81**	0.01***	1643.82	**1211.03**	0.01** [0.02**]
Real PCE	642.02	**173.32**	0.01***	789.10	**304.66**	0.01***	979.04	**686.76**	0.01** [0.02**]
Real Non-Res Fixed Inv	57.65	**40.80**	0.11 [0.12]	115.33	**78.73**	0.62 [0.63]	254.91	**235.59**	0.63
Real Res Fixed Inv	73.78	**25.20**	0.71 [0.72]	101.92	**53.79**	0.58	124.89	**98.07**	0.65 [0.67]
Real Federal C&GI	43.77	**14.23**	0.82	46.84	**22.36**	0.58	61.48	**47.24**	0.16
Real State/Local C&GI	135.51	**33.40**	0.01***	157.28	**58.33**	0.01***	269.73	**139.59**	0.01** [0.02**]
Real Change in Private Inv	30.23	**21.60**	0.44 [0.45]	47.05	**36.32**	0.21	53.89	**46.01**	0.73
Real Net Exports	167.51	**45.10**	0.27 [0.26]	263.69	**129.06**	0.26	**172.46**	195.93	0.48 [0.47]
Section 3: CPI and PCE Inflation									
CPI Inflation Rate	1.10	**1.07**	0.30 [0.29]	**0.74**	0.94	0.90	**0.65**	0.96	0.03**
Core CPI Inflation Rate	1.21	**0.72**	0.62	0.92	**0.71**	0.38	0.98	**0.82**	0.06*
PCE Inflation Rate	**0.68**	0.72	0.98 [0.99]	0.93	**0.91**	0.38 [0.37]	**0.61**	0.63	0.03**
Core PCE Inflation Rate	5.38	**5.23**	0.99	5.05	**4.89**	0.35	0.84	**0.65**	0.10 [0.09*]

Table B.14: Out-of-sample forecast accuracy (mean absolute errors) of best human and AI SPF forecasters

	Horizon 0			Horizon 1			Horizon 2		
Variable	**AI**	**Human**	**P-val**	**AI**	**Human**	**P-val**	**AI**	**Human**	**P-val**
Section 1: U.S. Business Indicators									
Nominal GDP	**0.23** [637.70]	308.70 [169.32]	0.12 [1.00]	**27.20** [797.10]	213.90 [40.26]	0.25 [0.50]	**85.45** [934.48]	107.12 [53.89]	0.51 [0.12]
GDP Price	**0.00** [0.73]	0.70 [0.29]	0.12 [0.38]	**0.17** [0.98]	0.66 [0.14]	0.25 [0.50]	**0.24** [0.97]	0.64 [0.13]	0.50 [0.62]
Corporate Profits	**276.12** [513.82]	283.06 [73.72]	0.13 [0.12]	**333.60** [387.50]	402.71 [153.01]	0.25	**456.30** [440.60]	457.83 [96.47]	0.51 [0.50]
Unemployment Rate	0.08	**0.07** [0.00]	0.24 [1.00]	0.18 [0.13]	**0.17** [0.05]	0.25	0.25 [0.12]	**0.20** [0.00]	1.00 [0.50]
Non-Farm Payroll	279.57 [238.89]	**125.76** [19.82]	0.13 [0.12]	302.94 [402.75]	**174.59** [18.57]	0.25 [0.12]	343.45 [390.21]	**246.47** [28.60]	0.50 [0.25]
Industrial Production	**0.27** [0.36]	0.41 [0.15]	0.12	**0.15** [0.24]	0.66 [0.07]	0.24 [0.25]	**0.30** [0.15]	0.80 [0.06]	0.49 [1.00]
Housing Starts	**0.03** [0.05]	**0.03** [0.00]	1.00 [0.50]	**0.06** [0.05]	0.07 [0.01]	0.25 [0.38]	0.07	**0.00** [0.01]	1.00 [0.50]
Treasury Bill Rate (3M)	**0.04** [0.21]	0.09 [0.00]	0.50 [0.12]	0.18 [0.36]	**0.17** [0.01]	0.25	0.32 [0.42]	**0.30** [0.03]	0.50 [0.12]
AAA Corp Bond Yield	**0.03** [0.35]	0.33 [0.00]	0.50 [0.12]	**0.21** [0.18]	0.23 [0.02]	0.50	0.10 [0.28]	**0.08** [0.02]	1.00 [0.12]
Treasury Bond Rate (10Y)	0.15 [0.07]	**0.14** [0.01]	0.87 [0.50]	**0.23** [0.06]	0.32 [0.04]	0.24 [0.75]	0.05 [0.08]	**0.00** [0.04]	1.00 [0.50]
Section 2: Real GDP and Its Components									
Real GDP	**0.07** [596.68]	118.32 [35.60]	0.12 [1.00]	**27.18** [675.05]	151.52 [28.95]	0.25 [0.12]	**163.80** [763.32]	170.62 [30.30]	0.49 [0.12]
Real PCE	**0.00** [423.04]	93.56 [20.87]	0.12	**30.48** [501.60]	125.42 [35.96]	0.25 [0.12]	**94.11** [575.48]	118.92 [71.89]	0.50 [0.25]
Real Non-Res Fixed Inv	**0.00** [8.20]	19.79 [4.13]	0.12 [1.00]	**28.95** [38.08]	39.50 [9.01]	0.25 [0.12]	98.68 [89.78]	**89.67** [27.90]	0.50 [0.12]
Real Res Fixed Inv	**0.00** [3.82]	2.32 [0.53]	0.13 [1.00]	**12.47** [11.08]	14.12 [1.69]	0.24 [0.12]	10.90 [22.18]	**9.77** [0.66]	0.50 [0.12]
Real Federal C&GI	**0.00** [5.70]	5.33 [0.08]	0.12 [1.00]	**3.87** [3.26]	5.97 [0.78]	0.25 [0.12]	**3.84** [10.12]	5.70 [0.78]	0.50 [0.75]
Real State/Local C&GI	**0.00** [14.52]	11.50 [2.09]	0.12 [1.00]	5.13 [20.90]	**2.11** [0.86]	0.25 [0.12]	19.48 [28.82]	**14.51** [1.15]	0.50 [0.12]
Real Change in Private Inv	**0.00** [9.95]	8.63 [0.17]	0.75 [1.00]	**16.52** [8.52]	16.61 [10.35]	0.24 [0.88]	**10.90** [55.00]	20.79 [22.37]	1.00 [0.12]
Real Net Exports	**0.00** [69.28]	11.15 [1.93]	0.13 [1.00]	**38.52** [173.79]	42.52 [48.44]	0.25 [0.12]	81.54 [183.46]	**78.10** [66.80]	0.50 [0.12]
Section 3: CPI and PCE Inflation									
CPI Inflation Rate	**0.93** [0.40]	1.00 [0.13]	0.25 [0.75]	**0.94** [0.55]	0.95 [0.20]	0.50 [0.12]	**1.18** [0.54]	**1.18** [0.27]	1.00 [0.38]
Core CPI Inflation Rate	0.67 [0.59]	**0.63** [0.17]	0.25 [0.12]	**0.43** [0.25]	0.61 [0.08]	0.50 [0.25]	**0.36** [0.22]	**0.36** [0.09]	1.00 [0.12]
PCE Inflation Rate	0.96 [0.46]	**0.85** [0.04]	0.51 [0.25]	0.80 [0.29]	**0.68** [0.17]	1.00 [0.25]	**0.24** [0.18]	0.63 [0.18]	1.00
Core PCE Inflation Rate	0.83 [0.50]	**0.61** [0.08]	0.50 [0.25]	**0.53** [0.19]	0.54 [0.10]	0.51 [0.12]	**0.00** [0.10]	0.08 [0.12]	1.00 [0.75]

Table D.15: Forecast accuracy (mean absolute errors) of AI-generated and AR(1) SPF forecasts relative to human accuracy

	Horizon 0			Horizon 1			Horizon 4		
Variable	**AI**	**AR(1)**	**P-val**	**AI**	**AR(1)**	**P-val**	**AI**	**AR(1)**	**P-val**
Section 1: US Business Indicators									
Nominal GDP	**0.24**	0.83	0.00***	**0.87**	1.25	0.00***	**1.14**	1.17	0.59
GDP Price	**1.00**	**1.00**	0.55	**1.00**	**1.00**	0.53	**0.99**	**0.99**	0.37
Corporate Profits	**0.65**	1.46	0.00***	**0.98**	1.38	0.00***	**0.98**	1.14	0.00***
Unemployment Rate	**1.41**	2.00	0.00***	**1.28**	1.51	0.00***	**1.12**	1.25	0.00***
Non-Farm Payroll	**2.13**	2.99	0.06*	**1.87**	1.93	0.94	1.50	**1.33**	0.01***
Industrial Production	**1.43**	2.23	0.01***	**1.19**	1.42	0.01**	**0.95**	1.20	0.00***
Housing Starts	**1.10**	1.41	0.00***	**1.08**	1.13	0.60	**1.01**	1.09	0.14
Treasury Bill Rate (3M)	**1.81**	2.06	0.00***	**1.54**	1.63	0.11	**1.06**	1.17	0.00***
AAA Corp Bond Yield	**1.48**	1.52	0.02**	**1.11**	1.17	0.00***	**0.97**	1.16	0.00***
BAA Corp Bond Yield	**1.54**	1.69	0.20	**1.16**	1.50	0.26	**0.99**	1.14	0.03**
Treasury Bond Rate (10Y)	**1.49**	1.56	0.04**	**1.12**	1.22	0.00***	**0.94**	1.16	0.00***
Section 2: Real GDP and Its Components									
Real GDP	**1.37**	2.10	0.18	**1.51**	1.89	0.19	**1.15**	1.26	0.28
Real PCE	**0.70**	1.68	0.00***	**1.30**	1.75	0.04**	**1.21**	1.23	0.68
Real Non-Res Fixed Inv	**0.81**	2.37	0.00***	**1.13**	1.88	0.01**	**0.98**	1.25	0.04**
Real Res Fixed Inv	**1.02**	1.88	0.00***	**1.43**	1.77	0.00***	**1.09**	1.14	0.15
Real Federal C&GI	**1.91**	5.36	0.05*	**1.56**	2.58	0.19	**1.27**	1.47	0.20
Real State/Local C&GI	**1.62**	3.80	0.02**	**1.30**	1.89	0.05*	**1.13**	1.24	0.22
Real Change in Private Inv	**0.57**	1.50	0.00***	**1.11**	1.19	0.54	1.16	**1.04**	0.09*
Real Net Exports	**1.26**	2.75	0.00***	**1.27**	1.57	0.02**	1.26	**1.20**	0.24
Section 3: CPI and PCE Inflation									
CPI Inflation Rate	1.20	**1.09**	0.10	1.05	**1.01**	0.26	**0.99**	1.01	0.71
Core CPI Inflation Rate	**1.09**	1.16	0.79	**0.92**	1.04	0.28	**1.02**	1.07	0.83
PCE Inflation Rate	**0.79**	1.12	0.00***	**0.96**	1.06	0.07*	**1.00**	1.01	0.58
Core PCE Inflation Rate	1.05	**0.99**	0.05*	1.07	**0.99**	0.02**	1.02	**0.98**	0.14